

Graph Convolutional Incomplete Multi-modal Hashing

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ABSTRACT

Multi-modal hashing (MMH) encodes multi-modal data into latent hash code, and has been widely applied for efficient large-scale multi-modal retrieval. In practice it is common that multi-modal data is often corrupted with missing modalities, e.g., social image often lacks its tags in image-text retrieval. Conventional MMHs can only learn on complete modalities, which however wastes a considerable amount of collected data. To fulfill this gap, this paper proposes Graph Convolutional Incomplete Multi-modal Hashing (GCIMH) to learn hash code on incomplete multi-modal data. GCIMH develops Graph Convolutional Autoencoder to reconstruct incomplete multi-modal data with effective exploit of its semantic structure. GCIMH further develops multi-modal and label networks to encode multiple modalities and label respectively. GCIMH can successfully transfer knowledge of autoencoder and label network to multi-modal hashing network using teacher-student learning framework. GCIMH can handle missing modalities in both offline training and online query stages. Extensive empirical studies on three benchmark datasets demonstrate the superiority of the proposed GCIMH over the state-of-the-arts on both complete and incomplete multi-modal retrieval.

CCS CONCEPTS

• Information systems → Multimedia and multimodal retrieval.

KEYWORDS

Hashing, Multi-modal, Graph convolutional network

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1 INTRODUCTION

Multi-modal retrieval [1, 24, 31] has gained increasing attention in recent years as heterogeneous data become ubiquitous. Multiple modalities describe the same object and provide both shared and supplementary information of the object [1]. Multi-modal retrieval aims to integrate multiple modalities to improve retrieval performance. In real-world application, real-valued multi-modal retrieval requires huge storage space and long retrieval time due to large volume and high dimension of multi-modal data.

Hashing [28] has been appreciated as an efficient solution to large-scale applications due to its low cost in terms of computation and storage. The basic idea of hashing [4, 28] is to learn a set of hash codes for high-dimensional data while preserving similarity structure. Benefiting from its advantages, hashing has been successfully applied to large-scale multi-modal retrieval. Multi-modal hashing (MMH) aims to fuse multi-modal data into latent hash code, by which efficient retrieval is performed. Existing MMHs can be divided into unsupervised [24, 31] and supervised methods [15, 16, 18, 19, 34] according to whether semantic supervision is used. Generally speaking, supervised MMH utilizes rich semantics to supervise learning of hashing model and hash code, and achieves better performance than unsupervised MMH.

Existing MMHs fuse multi-modal data by assuming that all modalities are available for each sample. This assumption greatly limits application of MMHs as in practice multi-modal data is often generated with missing modalities. For instance, in social image retrieval, social image often lacks its tags as users may not add textual description. Conventional MMHs can only learn on data with complete modalities, and cannot be directly applied to such incomplete multi-modal data. One possible solution is to only remain samples with complete modalities, however such scheme wastes a considerable amount of data with potentially useful information particularly in supervised setting where labeling is every time-consuming and expensive. To address the above issue, Adaptive Partial Multi-View Hashing (APMH) [35] is proposed to learn latent hash code by fusing incomplete multi-modal data. APMH employs multi-modal matrix factorization to learn shared and view-specific hash codes for complete and incomplete modalities respectively.

Generally speaking, APMH is a shallow hashing method, thus improved performance could be further expected. Therefore it still remains a research gap to develop deep multi-modal hashing that learns latent hash code for incomplete multi-modal retrieval.

To break restrict modality completeness requirement, we propose a novel deep multi-modal hashing method, i.e., Graph Convolutional Incomplete Multi-modal Hashing (GCIMH) to learn hash code by fusing incomplete multi-modal data. Inspired by great success of Graph Convolutional Network (GCN) [7, 9, 17, 37] on effective exploit of graph structure, GCIMH employs Graph Convolutional Autoencoder (GCAE) to reconstruct incomplete multi-modal data, and effectively imputes missing modalities using semantic neighbors. The proposed GCIMH comprises three components, i.e., GCAE, label network, and multi-modal hashing network, as illustrated in Figure 1. GCIMH employs teacher-student learning framework, where GCAE and label network are used to guide training of multi-modal hashing network. The main contributions of this work are as follows:

- We propose a novel deep multi-modal hashing method, i.e., Graph Convolutional Incomplete Multi-modal Hashing for fusing incomplete multi-modal data. To our knowledge, GCIMH is one of the first deep multi-modal hashing methods that can fuse incomplete multi-modal data for multi-modal retrieval.
- GCIMH develops Graph Convolutional Autoencoder to reconstruct incomplete multi-modal data, and fully exploits its semantic structure. It can effectively transfer the knowledge from autoencoder and label network to multi-modal hashing network using knowledge distillation. GCIMH can handle missing modalities in both training and query stages.
- Extensive empirical results on three benchmark datasets demonstrate that the proposed GCIMH obviously outperforms state-of-the-arts on both complete and incomplete multi-modal retrieval.

2 RELATED WORK

Multi-modal hashing (MMH) aims to fuse multiple modalities and learns latent hash code that can well discover correlation among multiple modalities. According to whether semantic information is used, existing MMHs are mainly divided into unsupervised MMHs [31], semi-supervised MMHs [36], and supervised MMHs [15, 16, 18, 19, 26]. Supervised MMHs employ rich semantics, e.g., label, pairwise semantic similarity to supervise hash code learning, and thus generally outperform unsupervised MMHs. For instance, Multiple Feature Kernel Hashing (MFKH) [16] learns hash function to preserve semantic similarity with linearly combined multiple kernels corresponding to different modalities. Flexible Online Multi-modal Hashing (FOMH) [18] learns hash code when multi-modal data arrives in streaming fashion and proposes an effective self-weighting scheme to capture variations of streaming data. Although supervised MMH achieves good retrieval performance, labeling is typically time-consuming and expensive. Semi-supervised MMH is a promising technique to address this problem by exploiting both labeled and unlabeled data. Efficient Semi-Supervised Multimodal Hashing (ESMH) [36] treats labeled and unlabeled data inequally and properly distinguishes importances of labeled and unlabeled hash codes. All the above MMHs require that data has complete

modalities, and cannot be directly applied to data with missing modalities. Therefore it still remains a research gap to effectively fuse incomplete multi-modal data and learn latent hash code for multi-modal retrieval.

Until now there have been several attempts [22, 25, 30, 32] of hashing towards handling incomplete multi-modal data. These methods are cross-modal hashing [2, 11, 12, 14, 33] that is proposed to support cross-modal retrieval, where query in one modality is used to retrieve samples in another modality. Multi-modal retrieval focused in this work typically requires data to be complete in both training and query stages, which is different from cross-modal retrieval. Adaptive Partial Multi-View Hashing (APMH) [35] is a MMH method that can handle incomplete multi-modal data in training stage. APMH learns shared and view-specific hash codes for complete and incomplete modalities respectively using multi-modal matrix factorization technique, and further exploits semantic structure. APMH is a shallow hashing method, thus improved performance could be further expected. In addition, FOMH [18] and FGCMH [20] can only handle incomplete multi-modal data in query stage. In this work, we propose deep multi-modal hashing to handle incomplete multi-modal data in both training and query stages.

3 THE PROPOSED METHOD

This section presents the details of the proposed Graph Convolutional Incomplete Multi-modal Hashing (GCIMH), including problem setup, formulation, and out-of-sample extension.

3.1 Problem Setup

This work focuses on two modalities, i.e., image and text modality, although the proposed GCIMH can be easily extended to multiple (more than two) modalities. Assume that we are given incomplete image and text multi-modal data that is denoted as $\mathbf{X} = [\mathbf{X}_c, \mathbf{X}_u] \in \mathbb{R}^{n_x \times d_x}$ and $\mathbf{Y} = [\mathbf{Y}_c, \mathbf{Y}_u] \in \mathbb{R}^{n_y \times d_y}$, where the n_c samples have complete modalities, i.e., $\mathbf{X}_c \in \mathbb{R}^{n_c \times d_x}$ and $\mathbf{Y}_c \in \mathbb{R}^{n_c \times d_y}$, while the n_{xu} samples have only image modality, i.e., $\mathbf{X}_u \in \mathbb{R}^{n_{xu} \times d_x}$ and the n_{yu} samples have only text modality, i.e., $\mathbf{Y}_u \in \mathbb{R}^{n_{yu} \times d_y}$, d_x and d_y are dimensions of image and text modality respectively, and we have $n_x = n_c + n_{xu}$, $n_y = n_c + n_{yu}$. In addition, its label is denoted as $\mathbf{L} = [\mathbf{L}_c, \mathbf{L}_{xu}, \mathbf{L}_{yu}] \in \{0, 1\}^{n \times c}$, where $n = n_c + n_{xu} + n_{yu}$, c is the number of classes. Moreover, $L_{ij} = 1$ if the j -th label is assigned to the i -th sample and $L_{ij} = 0$ otherwise. The incomplete image and text multi-modal data is illustrated in Figure 2 for easy understanding.

The goal of the proposed GCIMH is to learn deep multi-modal hashing model, and to fuse compact hash code $\mathbf{B} \in \{0, 1\}^{n \times d}$ on such incomplete multi-modal data, where d is the code length. After offline model training, GCIMH can support efficient and effective incomplete multi-modal retrieval in query stage, where query may have missing modality.

3.2 Formulation

The proposed GCIMH is illustrated in Figure 1. As shown in Figure 1, GCIMH comprises three components including Graph Convolutional Autoencoder, label network, and multi-modal hashing network.

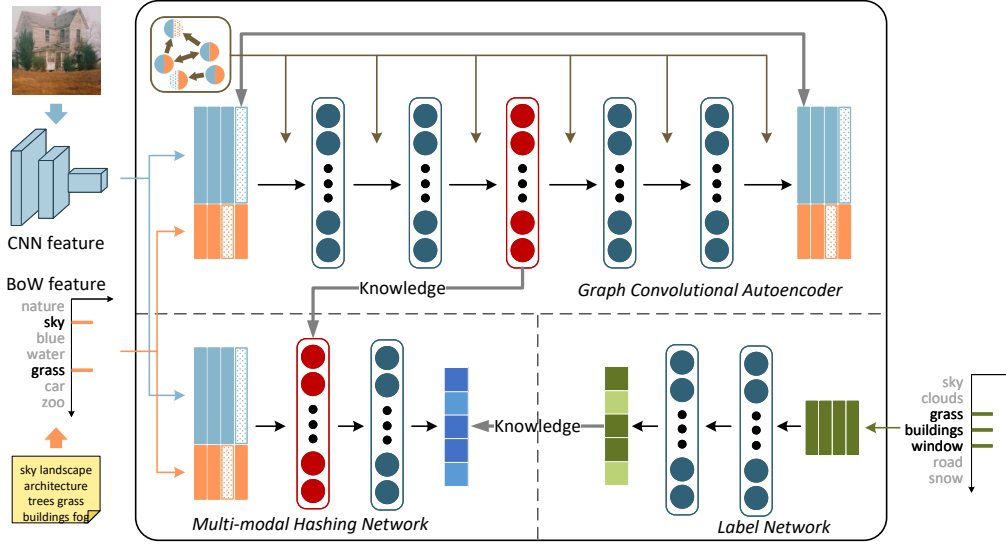


Figure 1: The illustration of the proposed GCIMH. The proposed GCIMH comprises three components including Graph Convolutional Autoencoder, label network, and multi-modal hashing network.

3.2.1 Graph Convolutional Autoencoder. This work proposes to employ Graph Convolutional Autoencoder (GCAE) [10] to reconstruct incomplete multi-modal data, and further learns latent feature. We first impute missing modality to construct complete multi-modal data. The missing modalities are first filled with mean features, i.e., $\bar{x}$ and $\bar{y}$ for image and text modality respectively, and the complete multi-modal data is thus denoted as $\tilde{X} = [X_c, X_u, \bar{X}] \in \mathbb{R}^{n \times d_x}$ and $\tilde{Y} = [Y_c, \bar{Y}, Y_u] \in \mathbb{R}^{n \times d_y}$, as shown in Figure 2. We then construct graph used in GCAE. A semantic matrix $W \in \mathbb{R}^{n \times n}$ is first defined to indicate whether two samples are similar, and $W_{ij} = L_i L_j^T$, L_i denotes label vector of the i -th sample. The aggregation of samples with missing modality leads to large reconstruction error, and thus these samples should not contribute to aggregation. With this idea, affinity graph $A \in \mathbb{R}^{n \times n}$ in this work is set as $A_{ij} = W_{ij} \zeta_j$, $\zeta_j = 1$ if modalities of j -th sample are complete and $\zeta_j = 0$ otherwise. The over-smoothing problem often occurs in Graph Convolutional Network (GCN) [9], which enables features of the nearby nodes to be more similar. To alleviate the above problem, we apply the following weighted scheme:

$$\tilde{A}_{ij} = \begin{cases} \left(A_{ij} / \sum_{j=1, i \neq j}^n A_{ij} \right) \times p, & i \neq j \\ 1 - p, & i = j \end{cases} \quad (1)$$

where $\tilde{A}$ is the weighted adjacency matrix, and p determines the weight assigned to a node itself and other nodes.

With the filled modalities and constructed graph, the concatenation of two modalities is fed into encoder network as follows: $T^G = f_E(\tilde{A}, [\gamma \tilde{X}, \tilde{Y}], \Theta_E)$, and γ reveals the relative importance of two modalities, $T^G \in \mathbb{R}^{n \times d_T}$ denotes latent feature, Θ_E is encoder network parameter. The latent feature is further fed into decoder

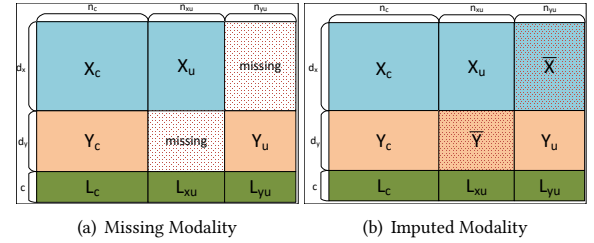


Figure 2: The illustration of multi-modal data. (a) The incomplete multi-modal data with missing modality, (b) The complete multi-modal data with imputed modality using mean feature.

network as follows: $[\gamma \hat{X}, \hat{Y}] = f_D(\tilde{A}, T^G, \Theta_D)$, and $\hat{X} \in \mathbb{R}^{n \times d_x}$ and $\hat{Y} \in \mathbb{R}^{n \times d_y}$ are reconstructed image and text modality respectively, and Θ_D is decoder network parameter. The layer-wise propagation rule of GCAE takes the following form:

$$H^{(l+1)} = \sigma(\tilde{A} H^{(l)} \Theta^{(l)}) \quad (2)$$

where $H^{(l)}$ and $H^{(l+1)}$ denote the input and output of the l -th layer convolution respectively, $\Theta^{(l)}$ is network parameter of the l -th layer in GCAE, and $\sigma(\cdot)$ is activation function.

The GCAE is optimized by minimizing reconstruction error over original data. The widely-used Mean Squared Error (MSE) and Binary Cross Entropy (BCE) are adopted for continuous image and discrete text modality respectively. In specific, GCIMH proposes

the following loss to train the above GCAE:

$$\mathcal{J}_G = - \sum_{i=1}^{n_y} \sum_{j=1}^{d_y} Y_{ij} \log(\hat{Y}_{ij}) + (1 - Y_{ij}) \log(1 - \hat{Y}_{ij}) \quad (3)$$

$$+ \lambda \sum_{i=1}^{n_x} \|\mathbf{X}_i - \hat{\mathbf{X}}_i\|_F^2$$

where the first and second terms are losses of text and image modality respectively, and λ balances the two reconstruction losses.

3.2.2 Label Network. GCIMH proposes label network to encode each label using embedding vectors by regarding label as an additional modality. The proposed label network is used to supervise training multi-modal hashing network such that learned hash code can well discover rich semantics. Specifically, GCIMH proposes label network $f_L(\mathbf{L}, \Theta_L)$ to encode label $\mathbf{L}$ as semantic feature $\mathbf{H}^L \in \mathbb{R}^{n \times d}$ in a common latent subspace shared by label, image, and text modalities, where Θ_L is label network parameter.

To this end, GCIMH aims to enable semantic feature $\mathbf{H}^L$ to well preserve semantic structure. GCIMH proposes the following loss to train label network:

$$\mathcal{J}_L = - \sum_{i,j=1}^n \left(S_{ij} \Omega_{ij}^L - \log(1 + e^{\Omega_{ij}^L}) \right) \quad (4)$$

where $\Omega_{ij}^L = \frac{1}{2} (\mathbf{H}_i^L)^\top (\mathbf{H}_j^L)$. Note that the above loss is the widely-used negative log likelihood loss [6] of within-modality semantic similarity.

3.2.3 Multi-modal Hashing Network. GCIMH proposes multi-modal hashing network to fully fuse the two modalities. It first concatenates image and text modality, and then feeds concatenation to network to obtain fused nonlinear feature as follows: $\mathbf{H}^F \in \mathbb{R}^{n \times d} = f_F([\gamma\tilde{\mathbf{X}}, \tilde{\mathbf{Y}}], \Theta_F)$, and f_F is multi-modal hashing network, Θ_F is its network parameter. In this work, inspired by the success of knowledge distillation [29], GCIMH proposes to transfer knowledge from other modules to hashing network to obtain high-quality hash code. Under teacher-student learning framework, GCIMH regards GCAE and label network as teacher modules, and hashing network as student module, and employs the two teachers to guide training of hashing network. Specifically, to transfer rich reconstruction knowledge of GCAE, GCIMH enables latent feature generated by hashing network to be close to that generated by GCAE. GCIMH simply minimizes the following loss:

$$\mathcal{J}_{FG} = \|\mathbf{T}^F - \mathbf{T}^G\|_F^2 \quad (5)$$

where $\mathbf{T}^F$ is feature of hidden layer in hashing network.

In addition, to transfer rich semantics of label modality, GCIMH aims to preserve image/text-label cross-modality similarity. To this end, the negative log likelihood loss is imposed on image/text-label pairs to train hashing network, and is defined as follows:

$$\mathcal{J}_{FL} = - \sum_{i,j=1}^n \left(S_{ij} \Omega_{ij}^F - \log(1 + e^{\Omega_{ij}^F}) \right) \quad (6)$$

where $\Omega_{ij}^F = \frac{1}{2} (\mathbf{H}_i^F)^\top (\mathbf{H}_j^F)$. In addition, it necessarily requires to binarize the network output to generate hash code. It is encouraging

Algorithm 1 Graph Convolutional Incomplete Multi-modal Hashing

Input: Image $\mathbf{X}$; text $\mathbf{Y}$; label $\mathbf{L}$; code length d ; parameters $\gamma, p, \lambda, \alpha, \beta$; mini-batch size m .

Output: network parameters $\Theta_L, \Theta_E, \Theta_D, \Theta_F$; hash code $\mathbf{B}$.

```

1: Initialize  $\Theta_L, \Theta_E, \Theta_D, \Theta_F$ ;
2: Impute missing modality to obtain  $\tilde{\mathbf{X}}$  and  $\tilde{\mathbf{Y}}$ ;
3: repeat
4:   for  $\lfloor \frac{m}{n} \rfloor$  iterations do ▷ Train Teacher Model
5:     Construct a mini-batch of image and text samples;
6:     Calculate reconstruction loss via (3) and update  $\Theta_E$  and  $\Theta_D$  by Adam algorithm;
7:   end for
8:   for  $\lfloor \frac{m}{n} \rfloor$  iterations do ▷ Train Teacher Model
9:     Construct a mini-batch of labels;
10:    Calculate label network loss via (4) and Update  $\Theta_L$  by Adam algorithm;
11:  end for
12:  for  $\lfloor \frac{m}{n} \rfloor$  iterations do ▷ Train Student Model
13:    Construct a mini-batch of image and text samples;
14:    Calculate hashing network loss via (8) by summing (5), (6), and (7), and update  $\mathbf{B}$  and  $\Theta_F$  by Adam algorithm;
15:  end for
16: until
```

to control quantization error by enabling the output to approach hash code by using the following quantization loss:

$$\mathcal{J}_{FQ} = \|\mathbf{H}^F - \mathbf{B}\|_F^2 \quad (7)$$

To this end, by summarizing the above three losses, i.e., $\mathcal{J}_{FG}$, $\mathcal{J}_{FL}$ and $\mathcal{J}_{FQ}$, we have the following objective function of multi-modal hashing network:

$$\mathcal{J}_F = \mathcal{J}_{FL} + \alpha \mathcal{J}_{FG} + \beta \mathcal{J}_{FQ} \quad (8)$$

where α and β are the two parameters to balance each term.

3.2.4 Optimization. The proposed GCIMH is a deep architecture that is trained under teacher-student learning framework. In this work, we first train two teacher models, i.e., GCAE and label network independently, and then train student model, i.e., multi-modal hashing network. The optimization procedure is repeated for several times. Note that the above procedure can achieve similar performance with end-to-end training. The training procedure of the proposed GCIMH is shown in Algorithm 1.

3.3 Out-of-Sample Extension

The proposed GCIMH has advantage of handling new query with complete or missing modalities. Specifically, given query with image modality, i.e., $\mathbf{x}_q$ and text modality, i.e., $\mathbf{y}_q$, its hash code can be generated by the learned multi-modal hashing network f_F as follows:

$$\mathbf{b}_q = \text{sign}(f_F([\gamma\mathbf{x}_q, \mathbf{y}_q], \Theta_F)) \quad (9)$$

In addition, given query with only image or text modality, GCIMH first fills missing modality with mean feature, and then generates hash code as follows:

$$\mathbf{b}_q = \text{sign}(f_F([\gamma\bar{\mathbf{x}}, \mathbf{y}_q], \Theta_F)) \text{ or } \text{sign}(f_F([\gamma\mathbf{x}_q, \bar{\mathbf{y}}], \Theta_F)) \quad (10)$$

Table 1: mAPs of all the MMHs on the three benchmark datasets. The bold and underline indicate best and second best respectively.

Method	MIRFLICKR-25K				MS COCO				NUS-WIDE			
	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
MFKH [16]	0.5864	0.6249	0.6501	0.6520	0.4728	0.4000	0.4734	0.5124	0.3875	0.4360	0.4669	0.4603
FOMH [18]	0.6863	0.7315	0.7428	0.7560	0.5319	0.6018	0.6276	0.6300	0.3974	0.5464	0.6278	0.6559
FGCMH [20]	0.7287	0.7236	0.7293	0.7334	0.4308	0.4488	0.4620	0.4735	0.4907	0.4674	0.4830	0.4817
FDMH [15]	0.7880	0.8059	0.8261	0.8283	<u>0.6278</u>	<u>0.6843</u>	<u>0.7164</u>	0.7412	<u>0.6861</u>	<u>0.6984</u>	<u>0.7228</u>	<u>0.7380</u>
SDMH [19]	0.7548	0.7874	0.7897	0.8059	0.5005	0.5506	0.5672	0.5755	0.6225	0.6433	0.6513	0.6592
ESuMH [36]	<u>0.8332</u>	<u>0.8429</u>	<u>0.8447</u>	<u>0.8484</u>	0.5401	0.5585	0.5924	0.6395	0.6640	0.6926	0.6985	0.6984
SAPMH [35]	0.7997	0.8151	0.8314	0.8330	0.5360	0.5668	0.5935	0.6004	0.6563	0.6732	0.6942	0.6887
GCIMH	0.8653	0.8770	0.8826	0.8855	0.6590	0.7007	0.7205	<u>0.7374</u>	0.7167	0.7328	0.7438	0.7483

4 EXPERIMENTS

This section evaluates the performance of the proposed method for multi-modal retrieval. The experiments are performed on a Ubuntu Enterprise 64-Bit Linux workstation with 128G memory and a NVIDIA A6000 GPU server.

4.1 Experimental Setup

4.1.1 Datasets. The experiment is conducted on three image-text benchmark datasets, i.e., MIRFLICKR-25K [5], MS COCO [13], and NUS-WIDE [3]. The multi-modal data is constructed to have image and text modality, where image modality is extracted by VGG19 Net [23] and text modality is represented by bag-of-words vector. The details of three benchmark datasets are as follows:

MIRFLICKR-25K [5] consists of 25,000 images collected from Flickr website. Each image is associated with several textual tags. Hence, each point is a image-text pair which is manually annotated with 24 unique labels. The image for each point is represented as a 4,096-dimensional VGG vector, and the text for each point is represented as a 1,386-dimensional bag-of-words vector. The 20,015 points that have at least 1 label and 1 textual tag are selected for experiments. The 2,234 samples are randomly selected as a query set and the rest 17,781 samples are used as database. The 5,000 samples are randomly selected from the database for training.

MS COCO [13] is constituted by two subsets of images: a training set with 82,783 training images and a validation set with 40,504 images, and each image is annotated with at least one of 80 classes. In our experiments, we merge the training images and validation images and remove those samples that have no text data. Finally, 122,218 image-text pairs are left for our experiments. The image for each point is represented as a 4,096-dimensional VGG vector, and the text for each point is represented as a 2,000-dimensional bag-of-words vector. The 5,000 samples are randomly selected as a query set and the rest samples are used as database. The 10,000 samples are randomly selected from the database for training.

NUS-WIDE [3] contains 260,648 web images and some images are associated with textual tags. Each image is annotated with one or multiple labels from 81 concept labels. We select 195,834 image-text pairs that belong to the 21 most frequent concepts. The image for each point is represented as a 4,096-dimensional VGG vector, and the text for each point is represented as a 1,000-dimensional bag-of-words vector. The 2,085 samples are randomly selected as a query

set and the rest 193,749 samples are used as database. The 10,000 samples are randomly selected from the database for training.

4.1.2 Experimental Setting. We compare the proposed method with seven state-of-art supervised multi-modal hashing methods, i.e., MFKH [16], FDMH [15], SDMH [19], ESuMH [36], FOMH [18], SAPMH [35], and FGCMH [20], where the first four can handle complete multi-modal data, and the remaining three can handle incomplete multi-modal data. Specifically, SAPMH can handle incomplete multi-modal data in training and query stages, and FGCMH and FOMH can only handle incomplete multi-modal data in query stage. For FGCMH and FOMH, multi-modal data with missing modalities are discarded in the training stage as the two methods do not empirically perform well using imputation scheme. The source codes of the comparison methods are kindly provided by the authors. For fair comparison, identical training and test sets apply to all the methods, and the parameters are set according to their papers.

For the proposed GCIMH, mini-batch size is set to 500, and max iteration number is set to 50. Adam optimizer [8] is utilized for optimization, and learning rate is varied in the range of $[10^{-3}, 10^{-1}]$. Among the parameters, γ is empirically set to 10, 0.5, 5 for MIRFLICKR-25K, MS COCO, and NUS-WIDE respectively; p , λ , α , β are empirically set to 0.6, 50, 0.6, 10^{-7} respectively. In GCAE, encoder and decoder are both GCNs with two 6,144-dimensional graph convolutional layers, and the dimension of latent feature is set to 8,192. In GCAE, ReLU is adopted as activation for hidden layers, and identity and Sigmoid functions are used to output reconstructed image and text features. The label network is built as a full-connected network with two 4,096-dimensional hidden layers. The multi-modal hashing network is built as a fully-connected network with one 8,192-dimensional and one 4,096-dimensional hidden layer. In hashing and label networks, ReLU is adopted as activation for hidden layers, and tanh activation is used to approximate hash code.

4.1.3 Evaluation Metric. We consider the widely-used metrics, i.e., mean Average Precision (mAP) and Precision-recall (PR) Curve to evaluate retrieval performance. The mAP is calculated using all the samples in database. In addition, we consider two multi-modal retrieval tasks, i.e., complete and incomplete multi-modal retrieval. For incomplete multi-modal retrieval, we evaluate performance when training set or query are incomplete.

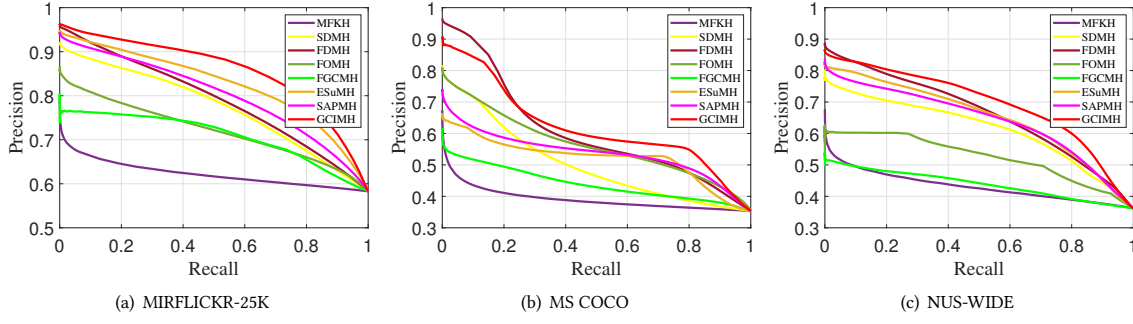


Figure 3: Precision-recall curves of all the MMHs on three benchmark datasets with respect to 32 bits.

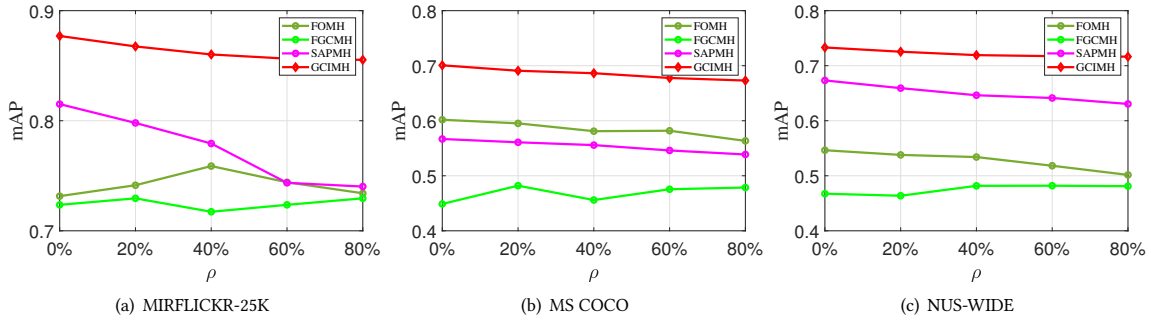


Figure 4: mAPs of the four incomplete MMHs with respect to different percentages of missing modality on the three benchmark datasets.

4.2 Evaluation on Complete Multi-modal Retrieval

This section evaluates performance of complete multi-modal retrieval, and compares the proposed GCIMH with seven state-of-art multi-modal hashing methods. The mAPs of all the methods are shown in Table 1, where bold and underline indicate the best and second best in each case. The PR curves of all the methods with respect to 32 bits are shown in Figure 3. From Table 1 and Figure 3, we can clearly observe that 1) the proposed GCIMH outperforms all the baselines in most cases. GCIMH has highest mAPs for 11 times among 12 cases, and it outperforms the best baselines averagely by 3.52%, 1.2%, and 2.35% on MIRFLICKR-25K, MS COCO, NUS-WIDE respectively. In addition, PR curves of GCIMH are generally above those of the baselines. 2) Among the four complete hashing baselines, FDMH and ESuMH have better performance, followed by SDMH and MFKH. 3) Among the three incomplete hashing baselines, SAPMH obviously outperforms FOMH and FGCIMH.

4.3 Evaluation on Incomplete Multi-modal Retrieval

This section evaluates performance of incomplete multi-modal retrieval, and compares the proposed GCIMH with the three incomplete hashing baselines, i.e., SAPMH, FGCIMH and FOMH.

4.3.1 Training Stage. We evaluate retrieval performance when training data is incomplete in training stage. To construct incomplete multi-modal data, we vary the percentage of training data with missing modality from the range of [0%, 20%, 40%, 60%, 80%], and conduct retrieval task using queries with complete modalities. The mAPs of the four incomplete MMHs with respect to different percentages denoted as ρ in Figure 4. From Figure 4, we can observe that the proposed GCIMH obviously outperforms the baselines among all the cases, indicating superiority of handling multi-modal data with different percentages of missing modality. In addition, as ρ increases, GCIMH achieves relatively stable performance, and the others have obvious performance drop. GCIMH benefits from effective reconstruction of missing modality in GCAE that considers aggregation of semantic neighbors. These empirical results imply that the proposed GCIMH can effectively handle training data with missing modality in training stage.

4.3.2 Query Stage. We further evaluate performance when query is incomplete in query stage. We train models using 40% percentage of incomplete training data and the remaining complete data. In query stage, we consider that query may have missing modality, and thus conduct retrieval task using either modality and two modalities of query. The mAPs of the four methods using image modality, text modality, and both modalities are shown in Table 2. The PR curves of all the methods with respect to 32 bits are shown in Figure 5, performances using text query is worse than those using

Table 2: mAPs of the four incomplete MMHs using different modalities of query in the three datasets. The bold indicates best.

Method	Modality	MIRFLICKR-25K				MS COCO				NUS-WIDE			
		16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits	16 bits	32 bits	64 bits	128 bits
FOMH [18]	I	0.7530	0.7817	0.7622	0.7394	0.4912	0.5269	0.6089	0.5990	0.4407	0.5116	0.6200	0.6487
	T	0.5903	0.6204	0.6167	0.6334	0.4605	0.5657	0.6186	0.5999	0.4456	0.4478	0.5191	0.5215
	I&T	0.7427	0.7588	0.7563	0.7642	0.4675	0.5810	0.6383	0.6231	0.4906	0.5340	0.6319	0.6631
FGCMH [20]	I	0.7273	0.7249	0.7269	0.7318	0.4692	0.4728	0.4874	0.4914	0.5016	0.4842	0.5405	0.5564
	T	0.5861	0.5882	0.5921	0.5916	0.4403	0.4532	0.4580	0.4539	0.3684	0.3652	0.3739	0.3782
	I&T	0.7182	0.7173	0.7234	0.7317	0.4384	0.4559	0.4734	0.4792	0.4053	0.4817	0.4626	0.4756
SAPMH [35]	I	0.7249	0.7771	0.7956	0.7851	0.5526	0.5585	0.5626	0.5910	0.6084	0.6221	0.6566	0.6623
	T	0.6573	0.6509	0.6618	0.6587	0.4748	0.5305	0.5228	0.4976	0.3794	0.3691	0.3762	0.3628
	I&T	0.7402	0.7793	0.7859	0.7862	0.5158	0.5557	0.5742	0.5883	0.6111	0.6463	0.6488	0.6641
GCIMH	I	0.7614	0.7744	0.7840	0.7762	0.6049	0.6248	0.6392	0.6466	0.6239	0.6450	0.6484	0.6601
	T	0.6864	0.6851	0.6850	0.6842	0.6176	0.6453	0.6585	0.6715	0.5445	0.5515	0.5509	0.5501
	I&T	0.8482	0.8602	0.8659	0.8658	0.6560	0.6863	0.7093	0.7115	0.7051	0.7192	0.7262	0.7304

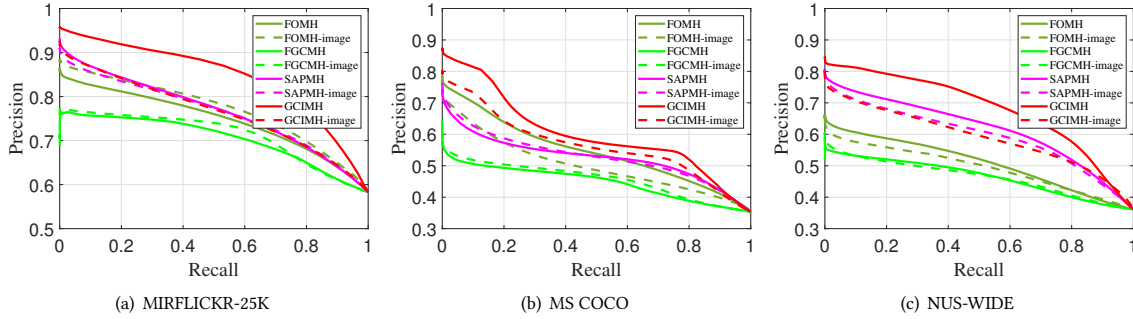
**Figure 5: Precision-recall curves of the four incomplete MMHs on three benchmark datasets with respect to 32 bits.**

image query, thus PR curves using text query are not plotted. From Table 2 and Figure 5, we see that GCIMH obviously outperforms the baselines in all the cases, indicating its superior performance in handling missing modality in query stage. In addition, mAPs using both modalities are higher than those using either modality, and mAPs using image modality as query are generally higher than those using text modality. These empirical results imply that the proposed GCIMH can effectively handle query with missing modality in query stage.

4.4 Further Analysis

4.4.1 Ablation Study. We discuss effectiveness of GCAE in the proposed GCIMH. GCIMH-A is a variation of GCIMH by removing GCAE module. The experiment is conducted on MS COCO, and 40% percentage of training data is incomplete, and all the queries are complete. The mAPs of the two methods are reported in Table 3, where we observe that GCIMH outperforms GCIMH-A, verifying effectiveness of GCAE.

4.4.2 Parameter Sensitivity. We analyze the sensitivity of the five important parameters in the proposed GCIMH, i.e., γ , p , λ , α , β . Specifically, γ reveals importances of each modality, p determines the weights of a node itself and other nodes in graph construction, and the other three parameters reveal importances of each term in loss function. The experiment is conducted on MS COCO with

Table 3: Ablation study of the proposed GCIMH. The bold indicates best.

Method	16 bits	32 bits	64 bits	128 bits
GCIMH-A	0.6485	0.6774	0.6875	0.7022
GCIMH	0.6560	0.6863	0.7093	0.7115

respect to 128 bits, and the mAPs of the proposed GCIMH with respect different parameters are shown in Figure 6. From Figure 6, it can be found that the performances remain relatively stable with the change of these parameters.

4.4.3 Visualization. We visualize learned hash code to qualitatively compare different hashing methods. The NUS-WIDE is adopted for experiment, and 20,000 samples that are annotated with only one label are randomly selected and code length is set to 32. The hash codes learned by seven methods are visualized into a 2-dimensional space with t-SNE [27], as illustrated in Figure 7. From Figure 7, we see that visualization is generally consistent with quantitative empirical results.

4.4.4 Convergence Analysis. We analyze convergence of the proposed GCIMH. Figure 8 shows the loss function values and mAPs of different epochs of the proposed GCIMH on MS COCO with respect to 64 bits. As can be seen, the loss function values first decrease

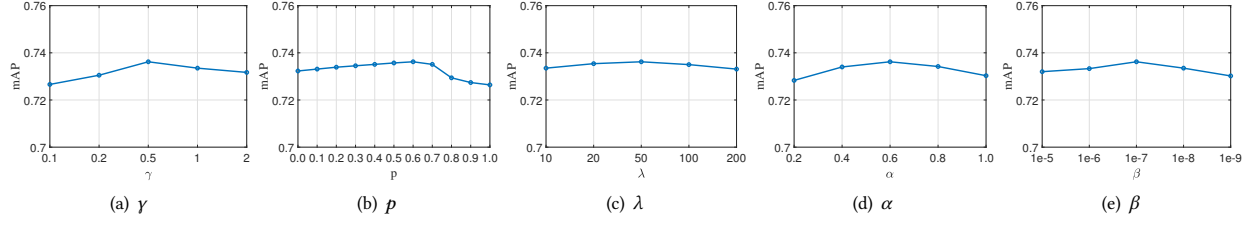


Figure 6: Parameter sensitivity analysis of the proposed GCIMH on MS COCO.

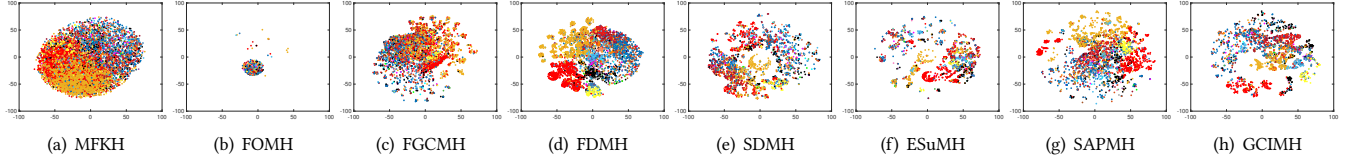


Figure 7: The t-SNE visualization of NUS-WIDE using all the MMHs.

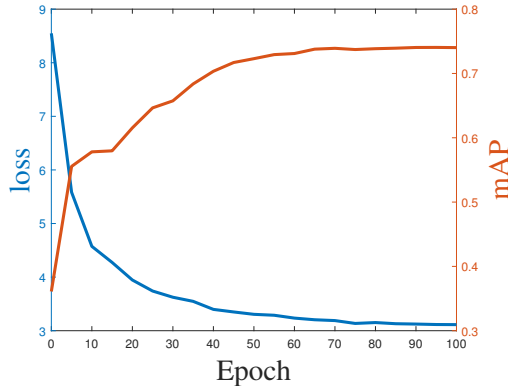


Figure 8: Loss function value curves and mAP curves of the proposed GCIMH on MS COCO with respect to 64 bits.

rapidly in the first 30 epochs, and then converge gradually within 70 epochs. Meanwhile, the mAPs first improve as epochs increase, and reach to highest values after 70 epochs.

4.4.5 Case Study. We present a case study on incomplete multi-modal retrieval, where the proposed GCIMH and the three incomplete hashing baselines, i.e., FOMH, FGCMH, and SAPMH are experimented on NUS-WIDE. Figure 9 illustrates the top-10 results of two randomly selected image and text queries, and all data is represented by image modality to enable results to be intuitive. A retrieved sample is marked green if it shares labels with query, and is marked red otherwise. From Figure 9, we can clearly observe that GCIMH can retrieve more correct samples than the three baselines.

5 CONCLUSION

This work paper studies the challenging hash code learning on incomplete multi-modal data, and proposes Graph Convolutional

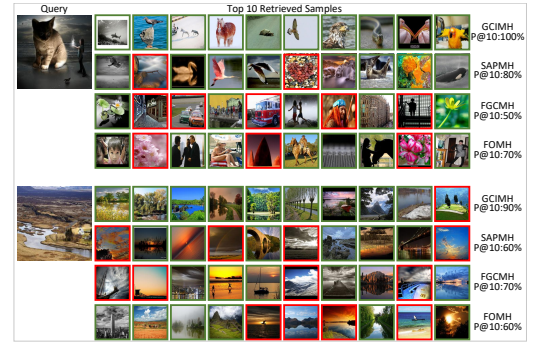


Figure 9: Top-10 retrieved results of the four incomplete MMHs on two randomly selected queries from NUS-WIDE.

Incomplete Multi-modal Hashing (GCIMH) for large-scale multi-modal retrieval. The key contribution is to employ Graph Convolutional Autoencoder to impute missing modalities using their semantic neighbors. GCIMH uses teacher-student learning scheme, and thus can handle missing modalities in both offline training and online query stages. The extensive empirical studies verify the theoretical findings, and demonstrate the proposed GCIMH works well on both complete and incomplete multi-modal retrieval.

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